Chapter 8. Canonical Correlation Analysis

Maximal correlation between two sets of variables

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1 Setting

We want to summarize the association between two sets of variables $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ with $\operatorname{Cov}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$, $\boldsymbol{\Sigma}_{11}, \boldsymbol{\Sigma}_{22} \succ 0$. Instead of inspecting pq correlations, maximize the correlation of linear combinations

$$\xi = \mathbf{g}^\top \mathbf{X}, \qquad \omega = \mathbf{h}^\top \mathbf{Y}.$$

For $q = 1$ this is the multiple correlation (Ch. 3). PCA maximizes variance *within* one set; CCA maximizes correlation *between* two sets.

Definition 8.1: canonical variates and canonical correlations

$$(\mathbf{g}_1, \mathbf{h}_1) = \arg \max_{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}),$$

$$(\mathbf{g}_k, \mathbf{h}_k) = \arg \max_{\substack{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q \\ \mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = 0,\ \mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}_j = 0,\ j=1,\dots,k-1}} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}).$$

$\xi_k = \mathbf{g}_k^\top \mathbf{X}$, $\omega_k = \mathbf{h}_k^\top \mathbf{Y}$ form the kth pair of canonical variates and $\rho_k = \operatorname{Corr}(\xi_k, \omega_k)$ is the kth canonical correlation. Correlation is scale-free, so normalize $\operatorname{Var}(\xi_k) = \operatorname{Var}(\omega_k) = 1$. The constraints mean $\operatorname{Cov}(\xi_k, \xi_j) = \operatorname{Cov}(\omega_k, \omega_j) = 0$, and $\rho_k \le \rho_j$ for $j < k$.

$\rho(\mathbf{X}, \mathbf{Y}) := \max \operatorname{Corr}(\mathbf{a}^\top \mathbf{X}, \mathbf{b}^\top \mathbf{Y}) = \rho_1$ is the **largest canonical coefficient**.

2 Solution via the Singular Value Decomposition

Motivation. The low-rank approximation problem is solved directly by the SVD (Eckart–Young), which also gives a numerically stable way to compute principal components.

Theorem 8.2: solution of CCA

Let $\mathbf{M} = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$ $(p \times q)$ have SVD $\mathbf{M} = \sum_{k=1}^{r} d_k \mathbf{a}_k \mathbf{b}_k^\top$, $d_1 \ge \cdots \ge d_r > 0$, $r = \operatorname{rank} \boldsymbol{\Sigma}_{12}$. Then

$$\rho_k = d_k, \qquad \mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2} \mathbf{a}_k, \qquad \mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2} \mathbf{b}_k.$$

Equivalently $\mathbf{a}_k$ is the kth eigenvector of $\mathbf{M}\mathbf{M}^\top = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1/2}$, $\mathbf{b}_k$ that of $\mathbf{M}^\top \mathbf{M} = \boldsymbol{\Sigma}_{22}^{-1/2} \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$, and ρ_k^2 is the eigenvalue.

Proof [Supplement]. Substituting $\mathbf{a} = \boldsymbol{\Sigma}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \boldsymbol{\Sigma}_{22}^{1/2} \mathbf{h}$,

$$\operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}) = \frac{\mathbf{g}^\top \boldsymbol{\Sigma}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}} \sqrt{\mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}}} = \frac{\mathbf{a}^\top \mathbf{M} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}} \sqrt{\mathbf{b}^\top \mathbf{b}}},$$

so we maximize $\mathbf{a}^\top \mathbf{M} \mathbf{b}$ over $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$. With $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, $\boldsymbol{\alpha} = \mathbf{U}^\top \mathbf{a}$, $\boldsymbol{\beta} = \mathbf{V}^\top \mathbf{b}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{b} = \sum_k d_k \alpha_k \beta_k \le d_1 \sum_k |\alpha_k \beta_k| \le d_1 \|\boldsymbol{\alpha}\| \|\boldsymbol{\beta}\| \le d_1,$$

with equality at $\mathbf{a} = \mathbf{a}_1$, $\mathbf{b} = \mathbf{b}_1$. Under $\mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = \mathbf{a}^\top \mathbf{a}_j = 0$ and $\mathbf{b}^\top \mathbf{b}_j = 0$ $(j < k)$ we have $\alpha_j = \beta_j = 0$ for $j < k$, and the same argument gives maximum d_k at $(\mathbf{a}_k, \mathbf{b}_k)$. ■

Theorem 8.3: properties of canonical variates

1. $\mathrm{Var}(\xi_k) = \mathrm{Var}(\omega_k) = 1$ and $\mathrm{Corr}(\xi_k, \xi_l) = \mathrm{Corr}(\omega_k, \omega_l) = 0$ for $k \neq l$.
2. $\mathrm{Corr}(\xi_k, \omega_l) = 0$ for $k \neq l$, $\mathrm{Corr}(\xi_k, \omega_k) = \rho_k$.
3. Invariance: canonical correlations are unchanged by $\mathbf{X} \mapsto \mathbf{D}_1\mathbf{X} + \mathbf{c}_1$, $\mathbf{Y} \mapsto \mathbf{D}_2\mathbf{Y} + \mathbf{c}_2$ with $\mathbf{D}_i$ non-singular; covariance and correlation matrices give the same ρ_k.
4. For $q = 1$, $\rho_1^2 = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$ is the squared multiple correlation.

Proof [Supplement]. (1) $\mathrm{Cov}(\xi_k, \xi_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{11}\mathbf{g}_l = \mathbf{a}_k^\top \mathbf{a}_l = \delta_{kl}$.

(2) $\mathrm{Cov}(\xi_k, \omega_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{12}\mathbf{h}_l = \mathbf{a}_k^\top \mathbf{M}\mathbf{b}_l = d_l \mathbf{a}_k^\top \mathbf{a}_l = d_l\delta_{kl}$.

(3) $\mathrm{Corr}(\mathbf{g}^\top\mathbf{D}_1\mathbf{X}, \mathbf{h}^\top\mathbf{D}_2\mathbf{Y}) = \mathrm{Corr}((\mathbf{D}_1^\top\mathbf{g})^\top\mathbf{X}, (\mathbf{D}_2^\top\mathbf{h})^\top\mathbf{Y})$ and $\mathbf{g} \mapsto \mathbf{D}_1^\top\mathbf{g}$ is a bijection, so the maxima agree.

(4) For $q = 1$ the only nonzero eigenvalue of $\mathbf{M}\mathbf{M}^\top$ is $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$. ■

3 Sample Canonical Correlations

Replace the covariances by $\mathbf{S}_{11} = \widehat{\mathrm{Cov}}(\mathbf{X})$, $\mathbf{S}_{22} = \widehat{\mathrm{Cov}}(\mathbf{Y})$, $\mathbf{S}_{12} = \widehat{\mathrm{Cov}}(\mathbf{X}, \mathbf{Y})$:

$$(\hat{\mathbf{g}}_k, \hat{\mathbf{h}}_k) = \arg \max_{\substack{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q \\ \mathbf{g}^\top\mathbf{S}_{11}\hat{\mathbf{g}}_j = 0,\ \mathbf{h}^\top\mathbf{S}_{22}\hat{\mathbf{h}}_j = 0,\ j=1,\dots,k-1}} \frac{\mathbf{g}^\top\mathbf{S}_{12}\mathbf{h}}{\sqrt{\mathbf{g}^\top\mathbf{S}_{11}\mathbf{g}\,\mathbf{h}^\top\mathbf{S}_{22}\mathbf{h}}}.$$

With $\mathbf{a} = \mathbf{S}_{11}^{1/2}\mathbf{g}$, $\mathbf{b} = \mathbf{S}_{22}^{1/2}\mathbf{h}$ this is maximizing $\dfrac{\mathbf{a}^\top\mathbf{S}_{11}^{-1/2}\mathbf{S}_{12}\mathbf{S}_{22}^{-1/2}\mathbf{b}}{\sqrt{\mathbf{a}^\top\mathbf{a}\,\mathbf{b}^\top\mathbf{b}}} = \max_{\|\mathbf{a}\|=\|\mathbf{b}\|=1} \mathbf{a}^\top\mathbf{M}\mathbf{b} = \mathbf{a}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top\mathbf{b}$, so

$$\mathbf{a}_i = \mathbf{v}_i(\mathbf{S}_{11}^{-1/2}\mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}\mathbf{S}_{11}^{-1/2}), \qquad \mathbf{b}_i = \mathbf{v}_i(\mathbf{S}_{22}^{-1/2}\mathbf{S}_{21}\mathbf{S}_{11}^{-1}\mathbf{S}_{12}\mathbf{S}_{22}^{-1/2}),$$

where $\mathbf{v}_i(\mathbf{M})$ is the ith eigenvector of $\mathbf{M}$.

3.1 Regularized CCA

If p or q is large relative to n, $\mathbf{S}_{11}, \mathbf{S}_{22}$ are singular or unstable and sample canonical correlations are biased towards 1. **Regularized CCA** adds a ridge term,

$$\mathbf{S}_{11} \mapsto \mathbf{S}_{11} + \lambda_1\mathbf{I}, \qquad \mathbf{S}_{22} \mapsto \mathbf{S}_{22} + \lambda_2\mathbf{I},$$

with λ_1, λ_2 chosen by cross-validation.

4 Testing Canonical Correlations

Under multivariate normality the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma}_{12} = \mathbf{0}$ (all $\rho_k = 0$) is

$$\Lambda^{2/n} = \frac{|\mathbf{S}|}{|\mathbf{S}_{11}||\mathbf{S}_{22}|} = \prod_{k=1}^{\min(p,q)} (1 - \hat{\rho}_k^2),$$

and Bartlett's approximation gives $-\big(n - 1 - \tfrac{p+q+1}{2}\big)\sum_k \ln(1-\hat{\rho}_k^2) \,\dot{\sim}\, \chi^2_{pq}$. The hypothesis that only the first k are nonzero, $H_0^{(k)} : \rho_{k+1} = \cdots = 0$, is tested sequentially with

$$-\Big(n - 1 - \frac{p+q+1}{2}\Big) \sum_{i=k+1}^{\min(p,q)} \ln(1-\hat{\rho}_i^2) \;\dot{\sim}\; \chi^2_{(p-k)(q-k)}.$$

Derivation: the determinant identity

By the block determinant formula, $|\mathbf{S}| = |\mathbf{S}_{22}||\mathbf{S}_{11} - \mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}| = |\mathbf{S}_{22}||\mathbf{S}_{11}|\,|\mathbf{I} - \hat{\mathbf{M}}\hat{\mathbf{M}}^\top|$, and the eigenvalues of $\hat{\mathbf{M}}\hat{\mathbf{M}}^\top$ are $\hat{\rho}_k^2$, so $|\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod(1 - \hat{\rho}_k^2)$.

5 Relations to PCA, Multiple Correlation and Discriminant Analysis

- **Multiple correlation**: CCA with $q = 1$.
- **Discriminant analysis**: with $\mathbf{Y}$ a set of group indicator (dummy) variables, the canonical variates of $\mathbf{X}$ give the same directions as Fisher's discriminants (Ch. 10).
- **PCA**: variance maximization within one set (eigendecomposition of $\boldsymbol{\Sigma}$) versus correlation maximization between two sets (SVD of $\boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}$).